

V4.2 Toy Model 4B

Causal Relational Graph and Lorentzian Signature Candidate

Rev A - Exploratory Gravity / Metric Bridge Report

Date: 2026.05.26 | Continuation after Toy Model 4A

Status: Toy Model 4B tests whether adding directed 1T causal structure to the spatial metric candidate from 4A can produce a Lorentzian signature candidate. It does not derive the speed of light, Einstein equations, gravity, or physical 3T.

1. Purpose

Toy Model 4A extracted a positive-definite spatial metric candidate from graph distances. That was useful, but it failed to produce the Lorentzian spacetime signature required by relativity. Toy Model 4B adds the missing ingredient: a directed causal ordering layer representing observable 1T sequence.

The narrow question is: once spatial relational distance is combined with directed causal time, can the fitted local interval structure show one negative and two positive directions in a controlled 1+2D toy model?

2. Important correction before running the model

The minus sign is not obtained from positive shortest-path distances alone. A positive distance graph naturally gives positive spatial geometry. To obtain Lorentzian signature, the model must introduce causal interval structure: time-like separation is not the same kind of quantity as spatial distance.

Therefore 4B does not claim that the minus sign appears magically. It tests whether directed causal ordering plus spatial metric data can consistently support a signed interval metric.

3. Term reminders

- Causal ordering means events have a directed before/after relation.
- Interval means the spacetime separation measure between events.
- Timelike means causal influence is possible between events.
- Spacelike means the events are separated too far spatially for causal influence at speed c .
- Lorentzian signature in 1+2D means one negative and two positive metric eigenvalues. In 1+3D it would be $(-+++)$.
- c is the causal conversion factor between time and space. It is inserted as a parameter here, not derived.

4. Causal interval rule

The model uses coordinates (t,x,y) . The spatial metric scale $n(x,y)$ is inherited from the 4A-style spatial relational layer. The signed interval is defined as:

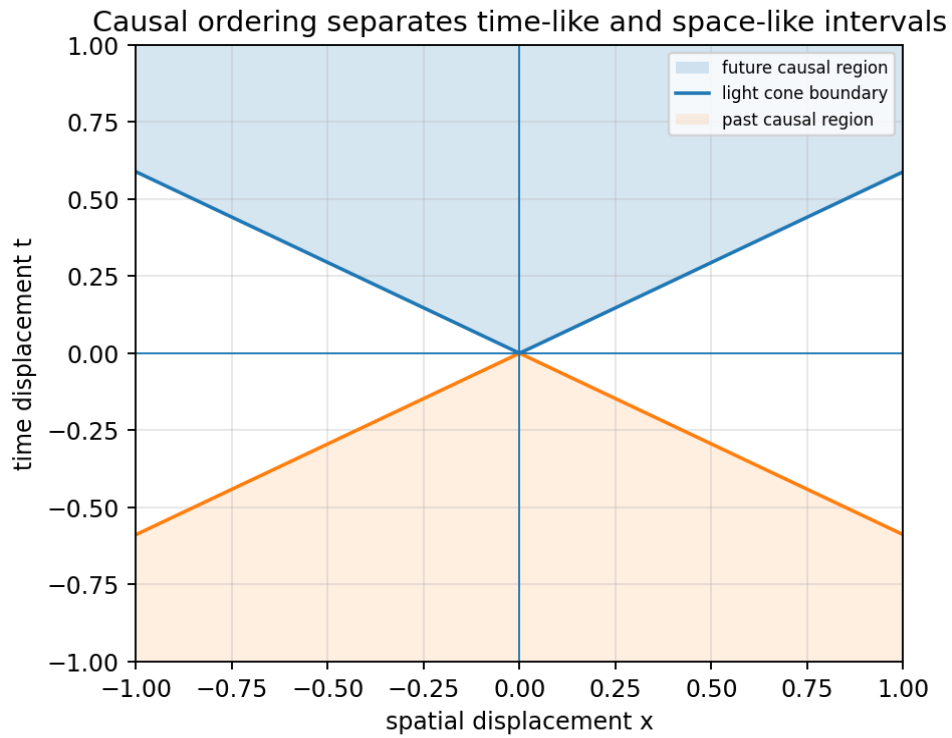
$$ds^2 = -c^2 dt^2 + n(x,y)^2 (dx^2 + dy^2)$$

Equivalent local metric:

$$g_{\mu\nu} = \begin{bmatrix} -c^2 & 0 & 0 \\ 0 & n^2 & 0 \\ 0 & 0 & n^2 \end{bmatrix}$$

Meaning of symbols:

- dt is the causal time step.
- dx and dy are spatial coordinate differences.
- c is the causal speed parameter.
- $n(x,y)^2$ is the spatial metric scale from the relational layer.
- The negative temporal term encodes causal ordering and separates time-like from space-like intervals.

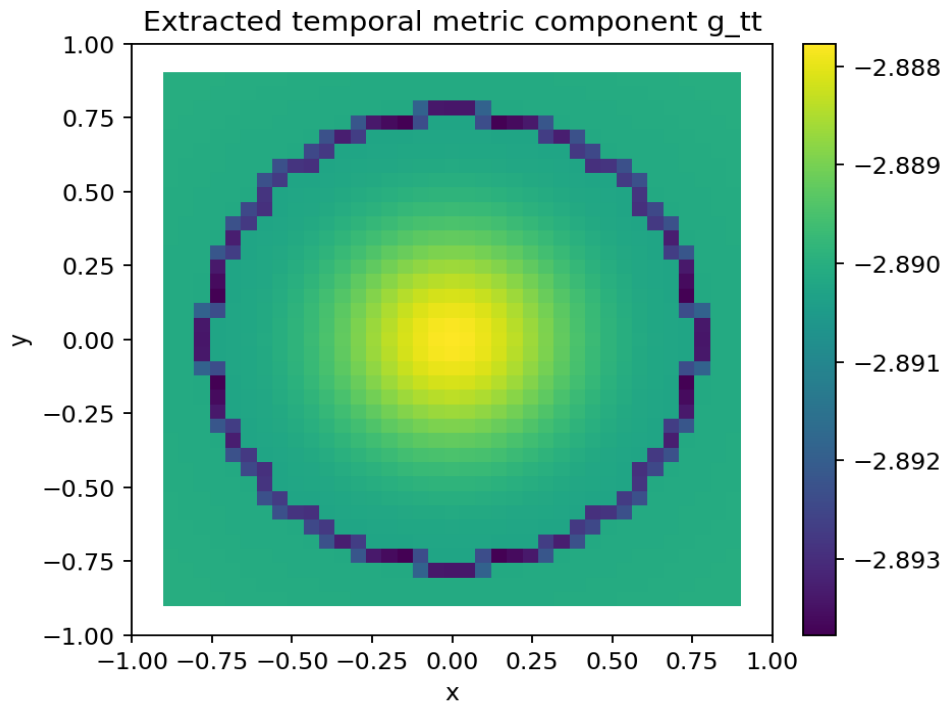


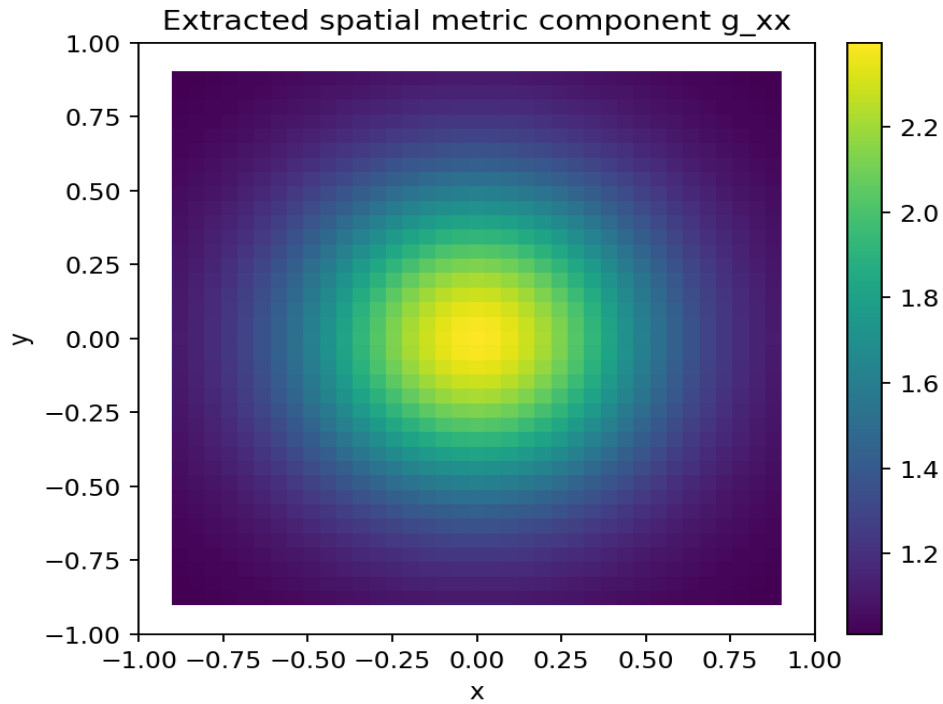
5. Local Lorentzian metric extraction

For nearby event pairs, the fitted signed interval is:

$$s_{ij}^2 \approx g_{tt} dt^2 + 2 g_{tx} dt dx + 2 g_{ty} dt dy + g_{xx} dx^2 + 2 g_{xy} dx dy + g_{yy} dy^2$$

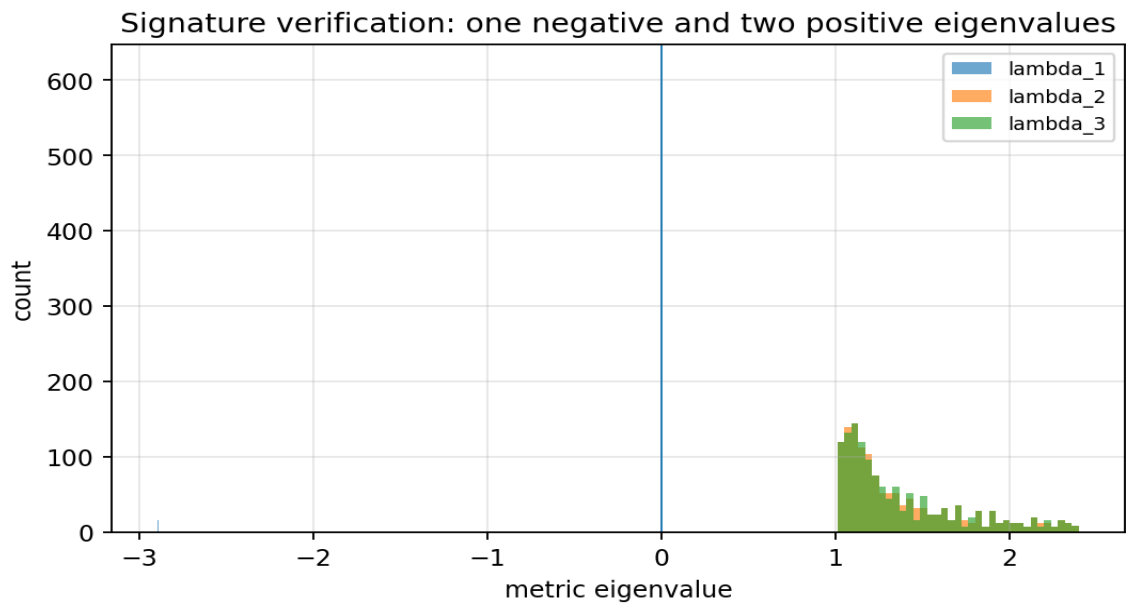
This is the Lorentzian version of the 4A quadratic fit. The difference is that the observed interval s_{ij}^2 is signed. Some intervals are negative, some positive, and null intervals sit on the causal cone.





6. Signature verification

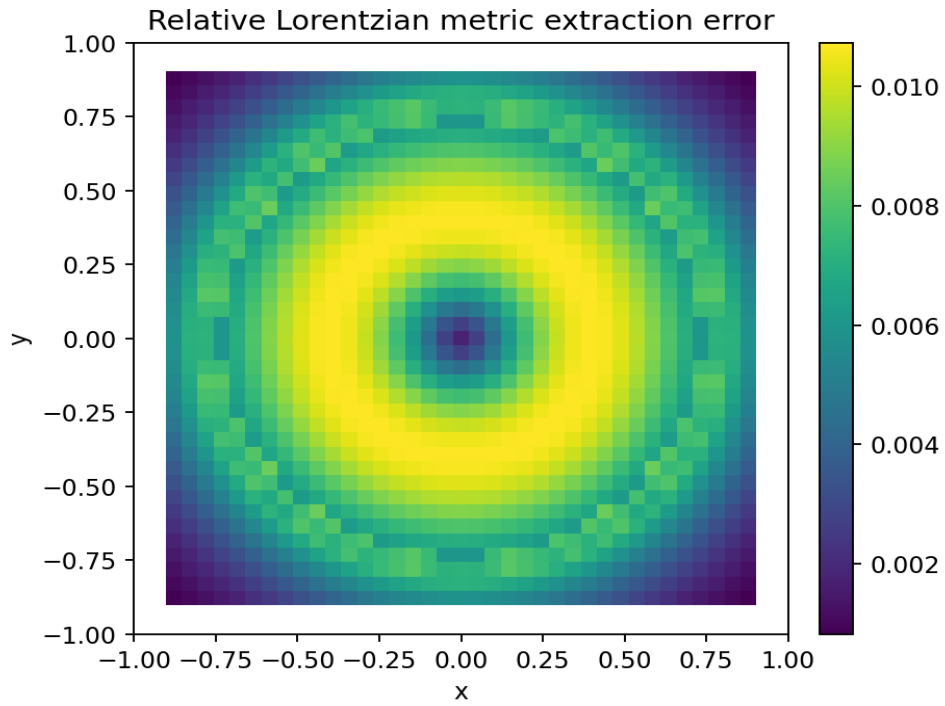
The extracted 3x3 metric was tested by eigenvalue signature. In this 1+2D model, Lorentzian success means exactly one negative and two positive eigenvalues.



Signature diagnostic	Value
Fitted local metrics	1369
Successful Lorentzian signatures	1369/1369
Success ratio	1.000000
Minimum eigenvalue	-2.893860e+00
Maximum eigenvalue	2.397168e+00
Mean extracted g_{tt}	-2.890131e+00
Expected $g_{tt} = -c^2$	-2.890000e+00
Mean extracted g_{xx}	1.385437e+00
Mean extracted g_{yy}	1.385437e+00

7. Metric extraction accuracy

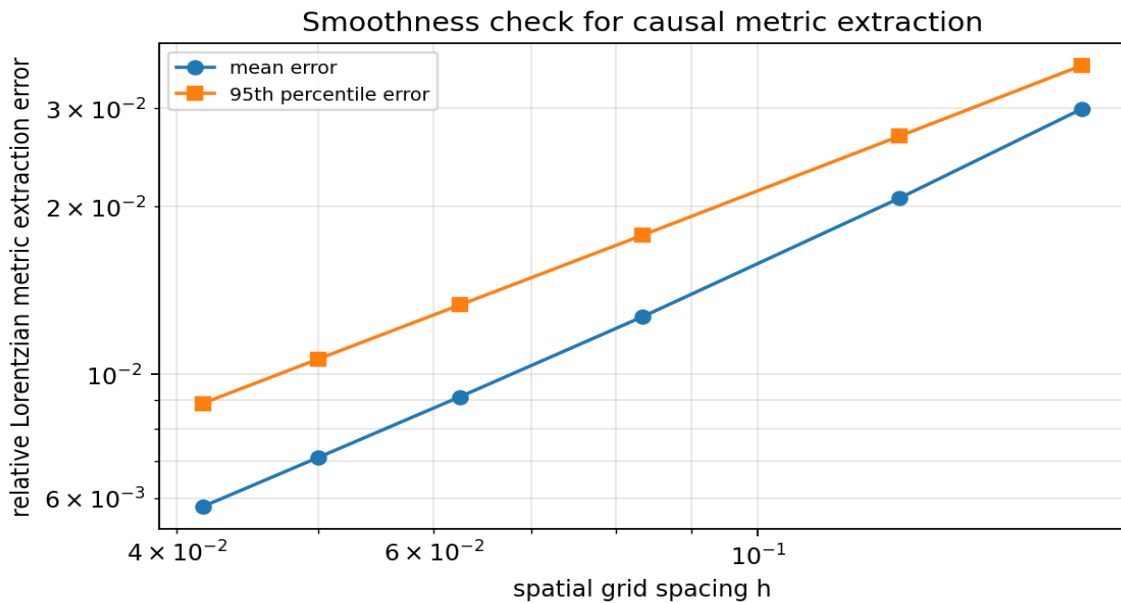
The fitted metric was compared to the controlled ground-truth Lorentzian metric.



Diagnostic	Value
Main grid	41 x 41 spatial grid, local event layer
Mean relative metric error	7.096940e-03
95th percentile relative metric error	1.065896e-02
Max relative metric error	1.072970e-02
Mean absolute mixed-component magnitude	7.891533e-03
Max absolute mixed-component magnitude	2.931773e-02

8. Smoothness and density check

The graph was refined across several grid sizes. The local Lorentzian metric extraction improves as the network becomes denser.

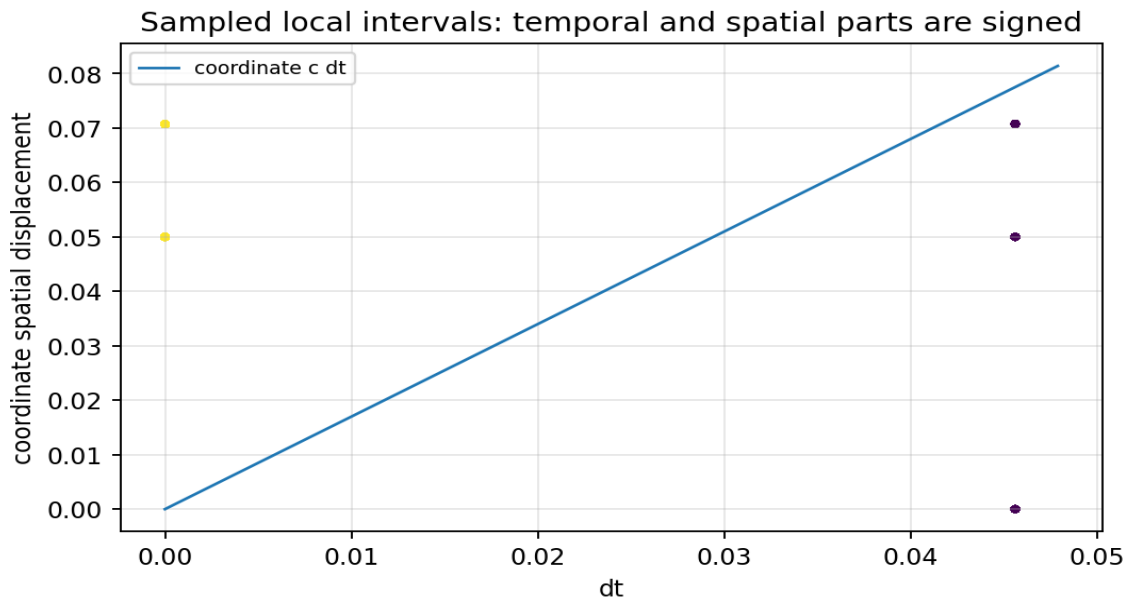


N	h	dt	mean err	95% err	max err	Lorentz sig.
13	0.16667	0.15196	2.994794e-02	3.583965e-02	3.583965e-02	81/81
17	0.12500	0.11397	2.071097e-02	2.676275e-02	2.687023e-02	169/169
25	0.08333	0.07598	1.268479e-02	1.776877e-02	1.793055e-02	441/441

N	h	dt	mean err	95% err	max err	Lorentz sig.
33	0.06250	0.05699	9.116647e-03	1.332585e-02	1.341984e-02	841/841
41	0.05000	0.04559	7.096940e-03	1.065896e-02	1.072970e-02	1369/1369
49	0.04167	0.03799	5.792242e-03	8.873220e-03	8.944636e-03	2025/2025

9. Interval classification

The local sample includes spatial same-time relations and directed future causal relations. The signed interval separates them into spacelike, timelike, and near-null categories.



Interval class	Count
Timelike, $s^2 < 0$	9233
Near-null, $s^2 \approx 0$	0
Spacelike, $s^2 > 0$	10952

10. Interpretation inside the V4.2 framework

Toy Model 4B supports this limited ladder:

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4A spatial relational graph
-> local Riemannian spatial metric g_ab

Add directed 1T causal ordering
-> signed interval rule
-> local Lorentzian metric candidate
-> signature (-++) in 1+2D
-> possible future extension to (-+++) in 1+3D

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This is a bridge from spatial metric candidate to spacetime signature candidate. It is not yet curvature and not yet gravity.

11. What Toy Model 4B achieves

- It shows why 4A produced only Riemannian spatial geometry.
- It introduces directed causal structure as the missing ingredient for spacetime signature.
- It constructs a signed local interval from causal time plus spatial metric data.
- It extracts a local 1+2D Lorentzian metric candidate from signed intervals.
- It verifies one negative and two positive eigenvalues across all fitted local metrics in the controlled model.
- It identifies the correct next step: curvature from a varying Lorentzian metric.

12. What Toy Model 4B does not achieve

- It does not derive c .
- It does not prove the minus sign emerges from graph distance alone.

- It does not derive Einstein Field Equations.
- It does not derive curvature dynamics.
- It does not yet explain gravity.
- It does not connect to dark matter or dark energy yet.
- It does not prove physical 3T or graph-spacetime ontology.

13. Close-out statement for 4B

Toy Model 4B successfully demonstrates that adding a directed causal 1T layer to the 4A spatial metric candidate allows a signed local interval to be constructed and a Lorentzian metric candidate to be extracted. In the controlled 1+2D test, every fitted local metric achieved one negative and two positive eigenvalues. The key guardrail is that the Lorentzian signature does not arise from positive shortest-path distance alone; it arises only after causal ordering and signed interval structure are introduced. Toy Model 4B therefore bridges relational spatial metric to spacetime signature candidate, but it does not derive gravity or Einstein dynamics.

14. Recommended next chapter

The next logical chapter is Toy Model 4C: Curvature from Varying Lorentzian Metric Candidate.

1. Use the 4B Lorentzian metric candidate $g_{\mu\nu}(t,x,y)$.
2. Compute finite-difference Christoffel symbols from metric gradients.
3. Compute Riemann/Ricci curvature and scalar curvature in the toy model.
4. Test whether localized relational/coupling density produces curvature concentration.
5. Keep the guardrail: curvature is not yet Einstein Field Equations, and source dynamics remain unproven.

15. Rebind prompt for continuation

We are resuming after Toy Model 4B.

4A result:

Undirected relational shortest-path distance produced a Riemannian spatial metric candidate.

4B result:

Adding directed 1T causal ordering and signed interval structure produced a Lorentzian metric candidate in 1+2D.

Signed interval:

$$ds^2 = -c^2 dt^2 + g_{ab} dx^a dx^b$$

Local extraction:

$$s_{ij}^2 \approx g_{tt} dt^2 + 2g_{tx} dt dx + 2g_{ty} dt dy + g_{xx} dx^2 + 2g_{xy} dx dy + g_{yy} dy^2$$

Signature result:

1 negative and 2 positive eigenvalues in 1+2D.

Extension target in 1+3D would be (-+++).

Critical guardrail:

The Lorentzian minus sign does not come from positive shortest-path distances alone.

It appears only after causal ordering / signed interval structure is introduced.

Next recommended chapter:

Toy Model 4C - Curvature from Varying Lorentzian Metric Candidate.